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Advanced Natural Language Processing

Lecture 22: Multimodal Large Language Model

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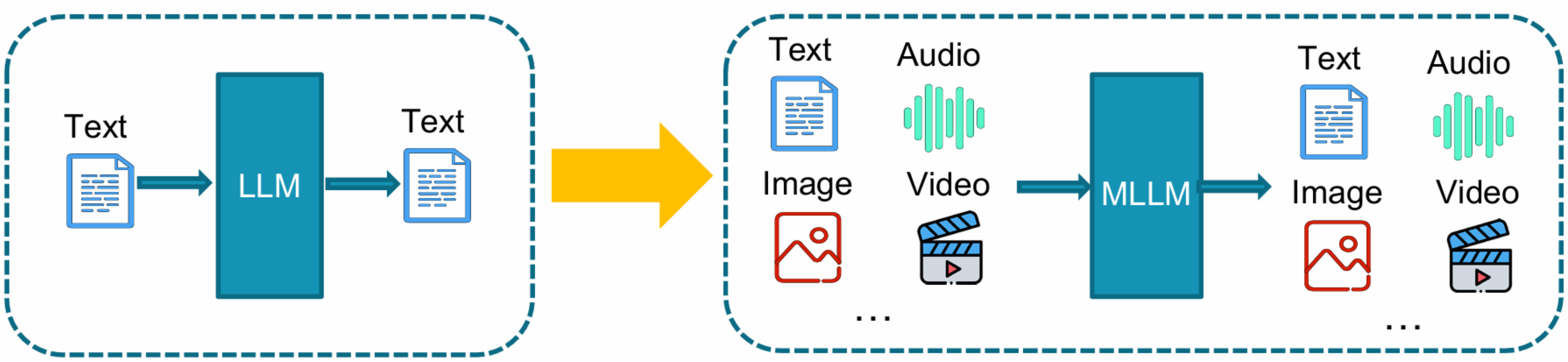
Content

Large Language Models (LLMs)

- **What LLMs Excel At:**
  1. Understanding and generating human language.
  2. Reasoning over textual information.
  3. Translation, summarization, Q&A.

- **BUT,** The world is **NOT JUST** text.



Why Do We Need Multimodality?

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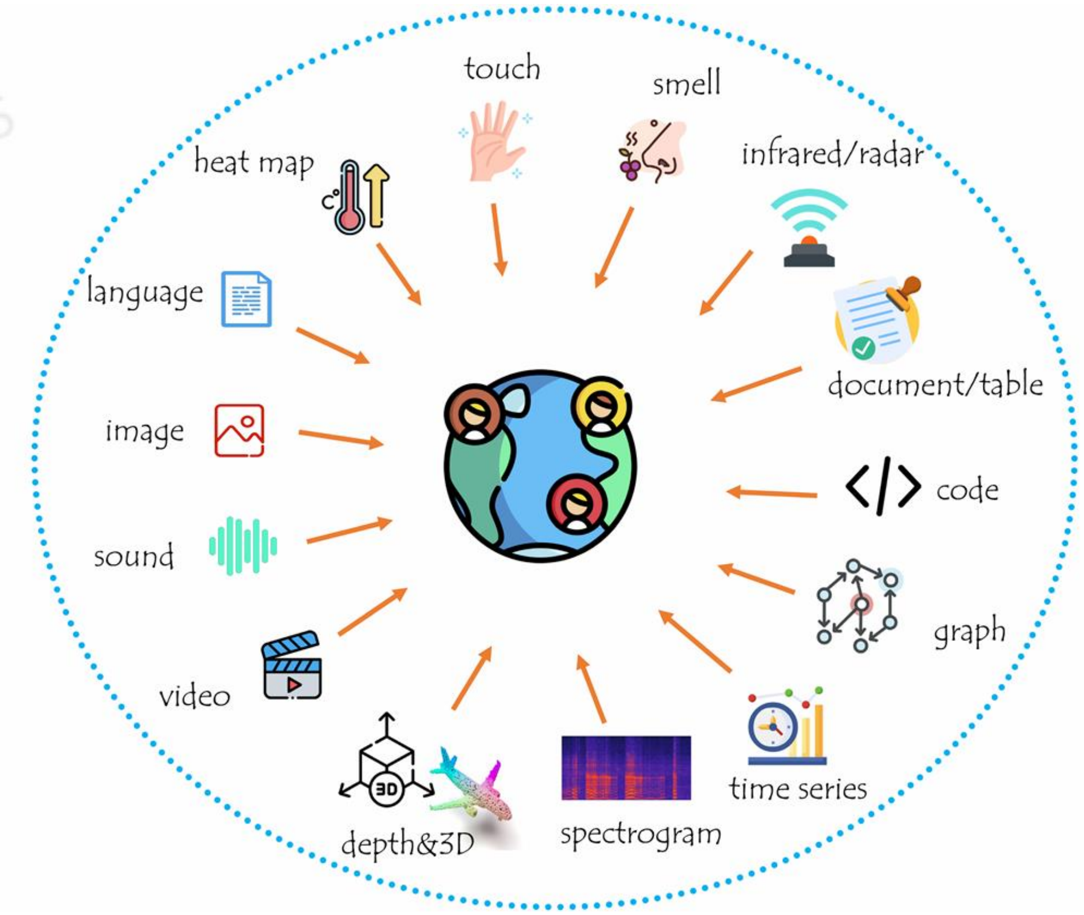
- **Human Cognition is Multimodal:** We learn and reason by combining senses (seeing, hearing, reading).
- **Richer Context:** Modalities provide complementary information. An image can convey what a thousand words cannot.
- **Grounding Language:** Connecting language to the physical world. "Red" is not just a word; it's a visual experience.



Trends of MLLMs

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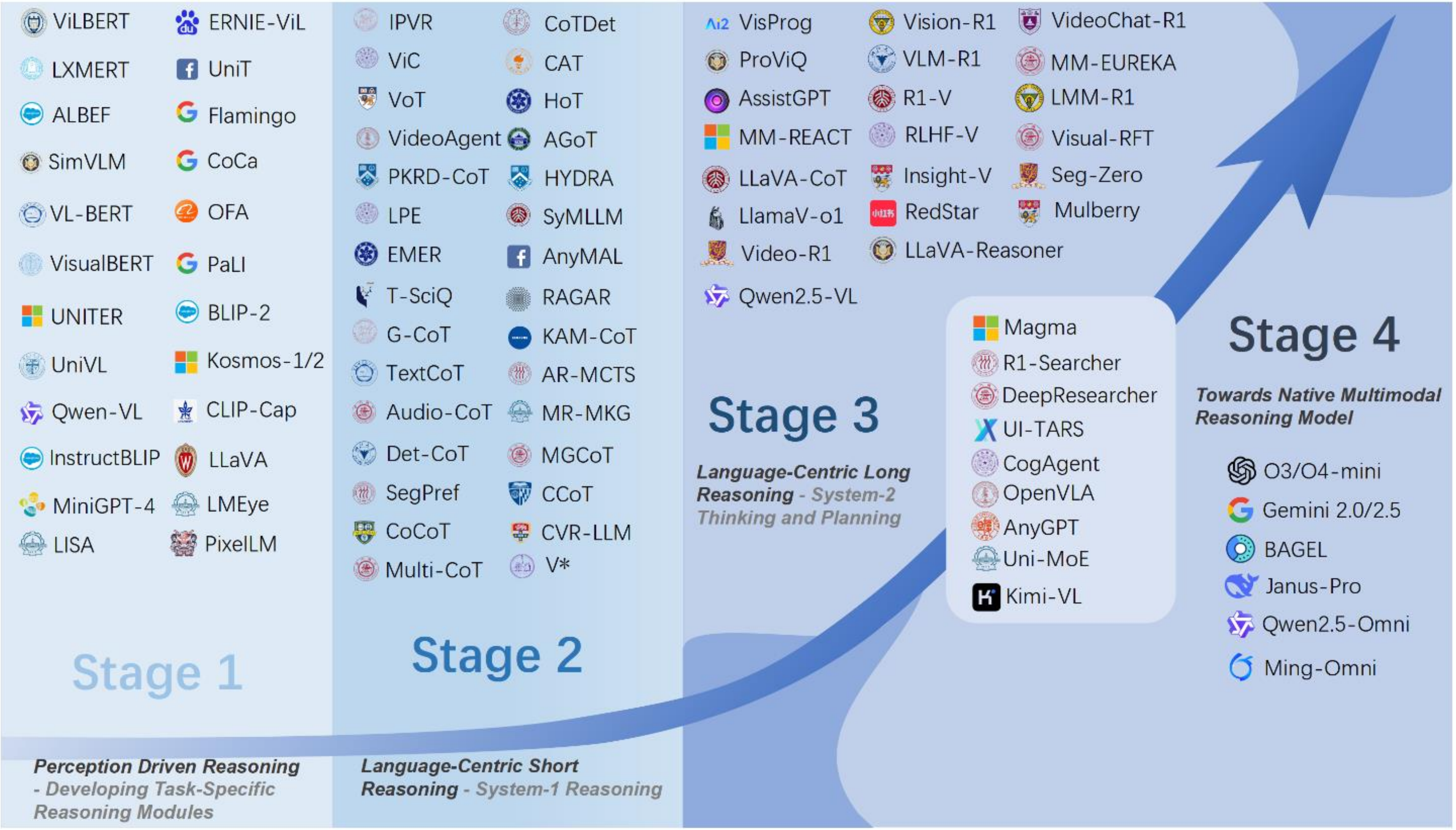
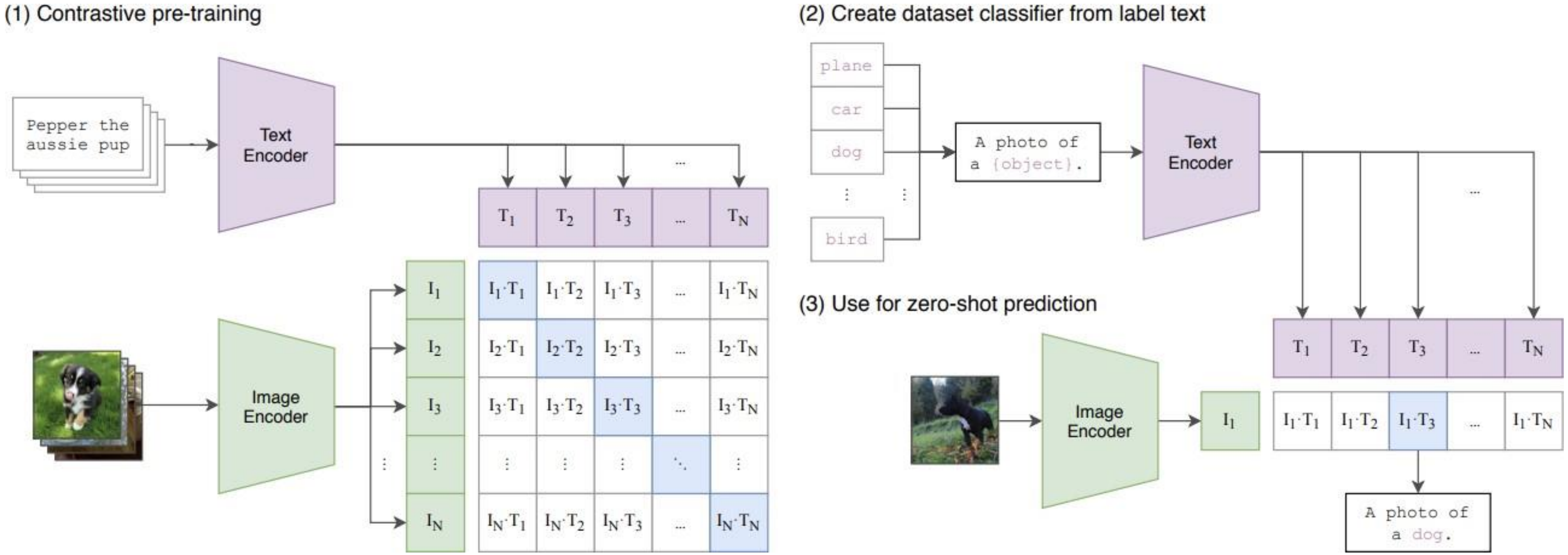


CLIP

Learning image representations from web-scale noisy text supervision

- Training: simple contrastive learning, and the beauty lies in large-scale pre-training
- Downstream: zero-shot image classification and image-text retrieval
- Image classification can be reformatted as a retrieval task via considering the semantics behind label names



1 Learning transferable visual models from natural language supervision, ICML2021
2 Scaling up visual and vision-language representation learning with noisy text supervision, ICML2021

CLIP

Trained with contrastive loss

- Each image can be paired with N possible texts, and the model tries to predict the correct one

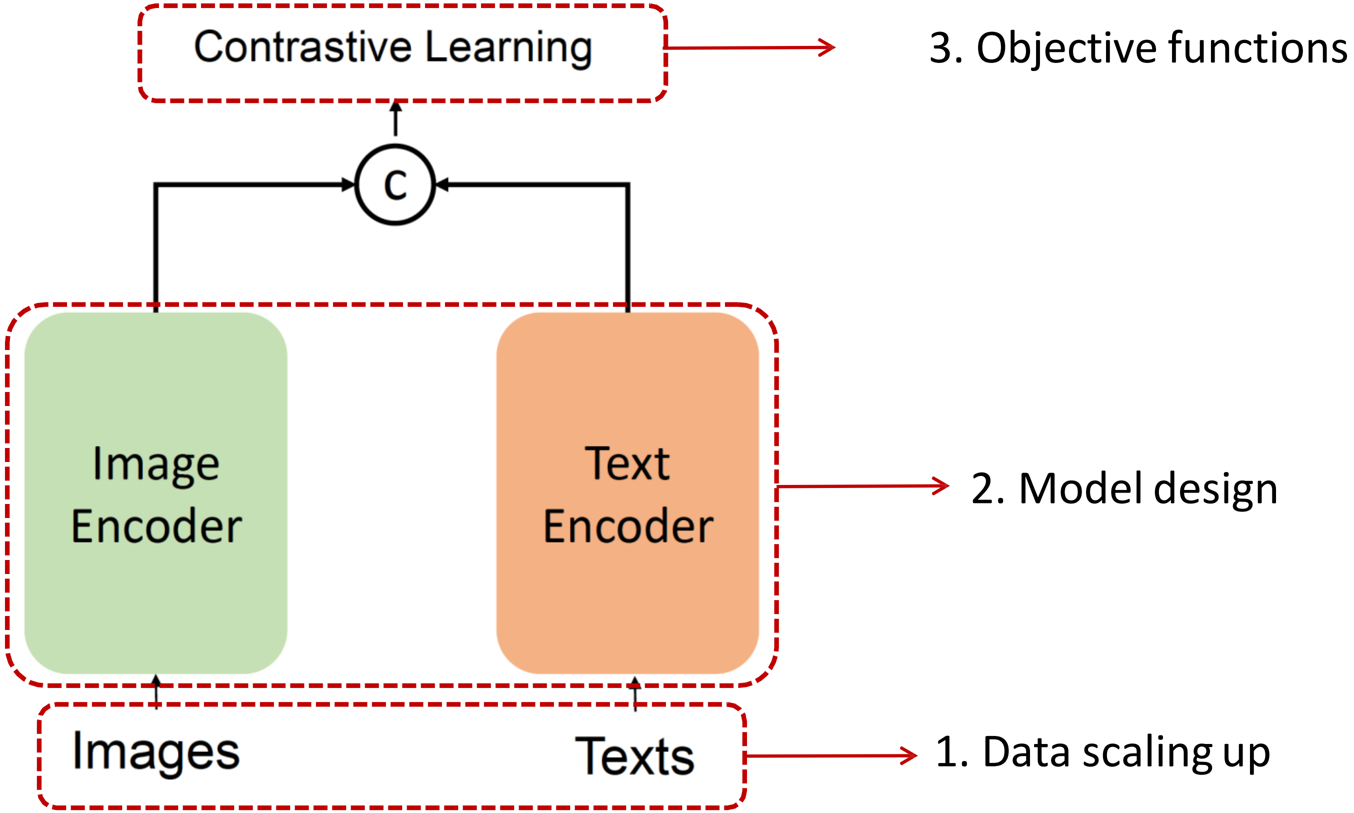
$$L_{\text{contrastive:txt2im}} = -\frac{1}{N}\sum_{i}^{N}\log\left(\frac{\exp(L_i^T V_i \beta)}{\sum_j^N \exp(L_i^T V_j \beta)}\right)$$

- Each text can be paired with N possible images, and the model tries to predict the correct image

$$L_{\text{contrastive:im2txt}} = -\frac{1}{N}\sum_{i}^{N}\log\left(\frac{\exp(V_i^T L_i \beta)}{\sum_j^N \exp(V_i^T L_j \beta)}\right)$$

How to Improve CLIP

- Since the birth of CLIP, tons of follow-up works and applications



Data Scaling Up

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- Reproducible scaling laws for CLIP training
  - Open large-scale LAION-2B dataset
  - Pre-training OpenCLIP across various scales

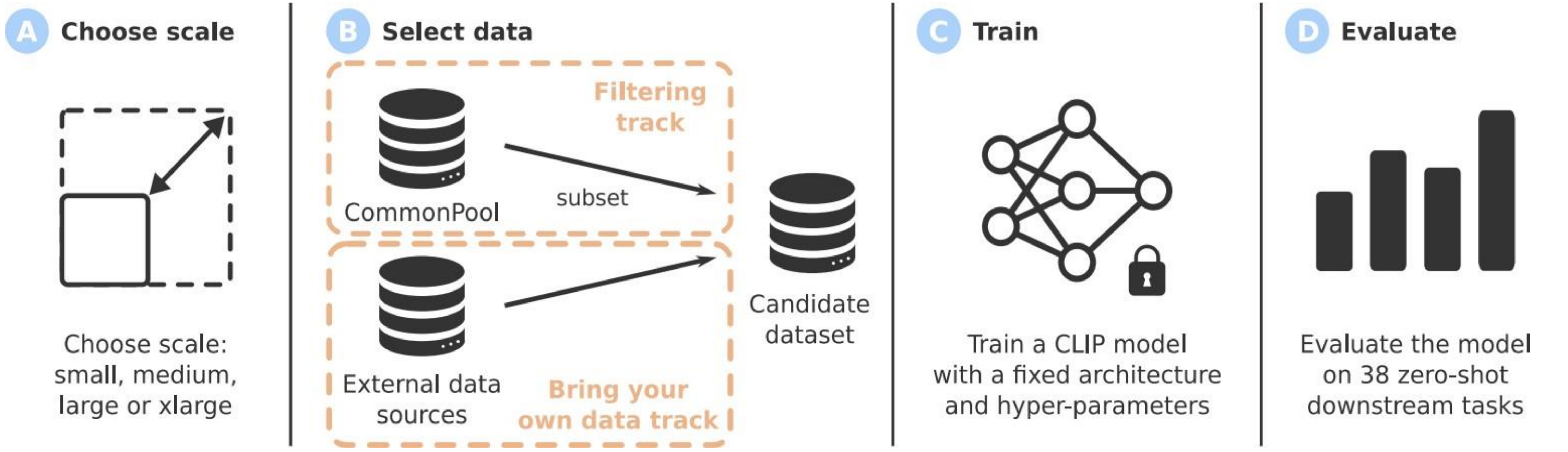
	Data	Arch.	ImageNet	VTAB+	COCO
CLIP [55]	WIT-400M	L/14	75.5	55.8	61.1
Ours	LAION-2B	L/14	75.2	54.6	71.1
Ours	LAION-2B	H/14	78.0	56.4	73.4

- DataComp: In search of the next-generation image-text datasets
  - Instead of fixing the dataset, and designing different algorithms,
  - Authors propose to fix the CLIP training method, but select the datasets instead
  - A benchmark used to design new filtering techniques or curate new data sources.
  - Evaluate the new dataset by running standardized CLIP training code and testing the resulting model on 38 downstream test sets

1 Reproducible scaling laws for contrastive language-image learning, CVPR 2023

2 Datacomp: In search of the next generation of multimodal datasets, 2023

Data Scaling Up



1 Reproducible scaling laws for contrastive language-image learning, CVPR 2023
2 Datacomp: In search of the next generation of multimodal datasets, 2023

CLIP: Pseudocode

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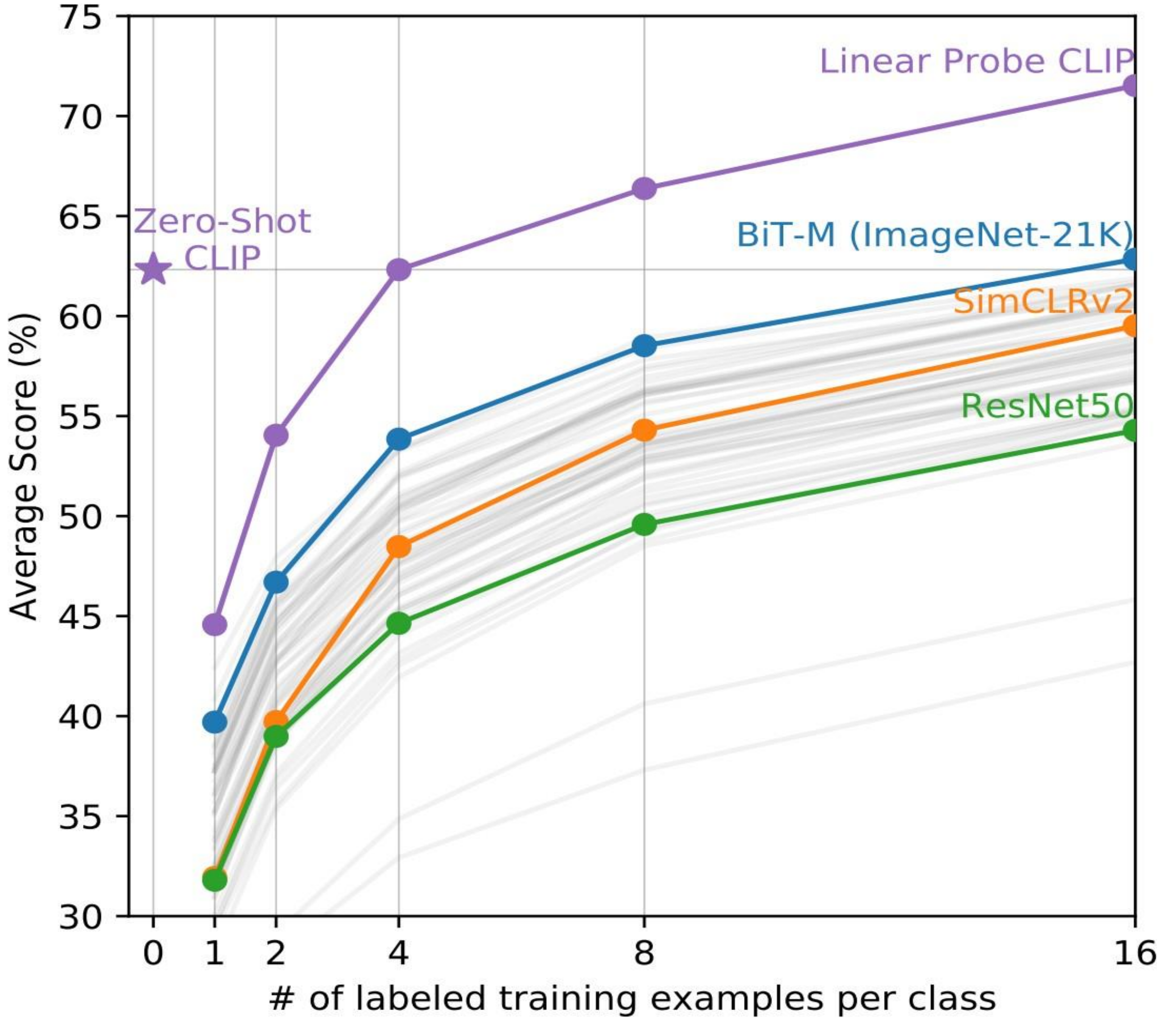
```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

open_clip/loss.py code

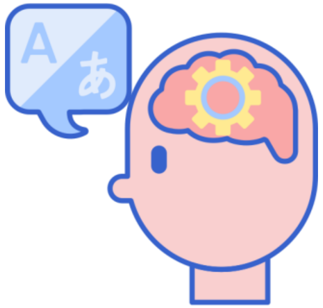
Few-shot Performance

- Zero-shot CLIP outperforms other few-shot baselines
- Few-shot CLIP further improves with a few labeled data.



Overview of MLLM Architecture

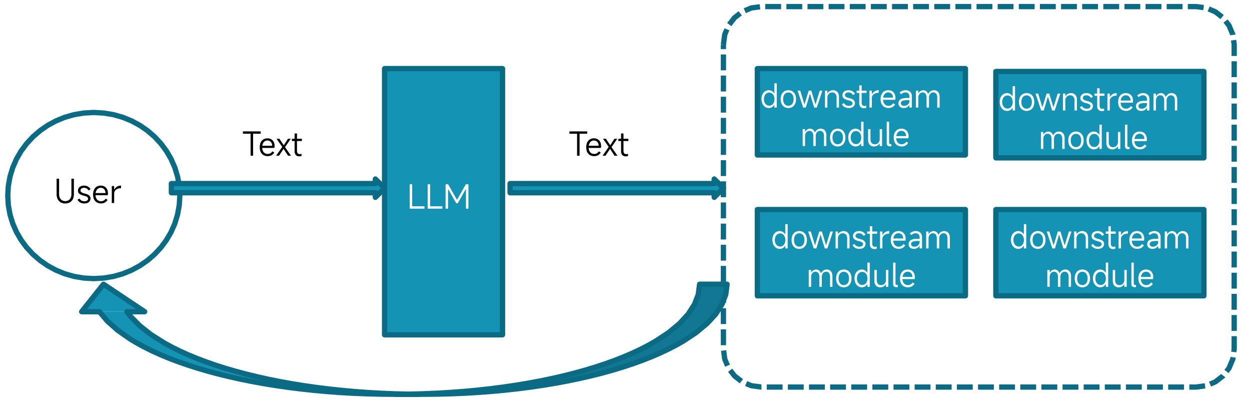
- Preliminary Idea: Intelligence over Language (Language-centric MLLM)
  - Emergent phenomena have extensively already occurred in language-based LLMs.
  - These LLMs now generally possess very powerful semantic understanding capabilities. This also implies that language is a crucial modality for carrying intelligence.



language

Overview of MLLM Architecture

- Architecture-I: LLM as Discrete Scheduler/Controller
  - The role of the LLM is to receive textual signals and instruct textual commands to call downstream modules.
- Key feature:
  - All message passing within the system, such as "multimodal encoder to the LLM" or "LLM to downstream modules", is facilitated through pure textual commands as the medium.



Examples:

- Visual-ChatGPT
- HuggingGPT
- MM-REACT
- ViperGPT
- AudioGPT
- LLaVA-Plus

Overview of MLLM Architecture

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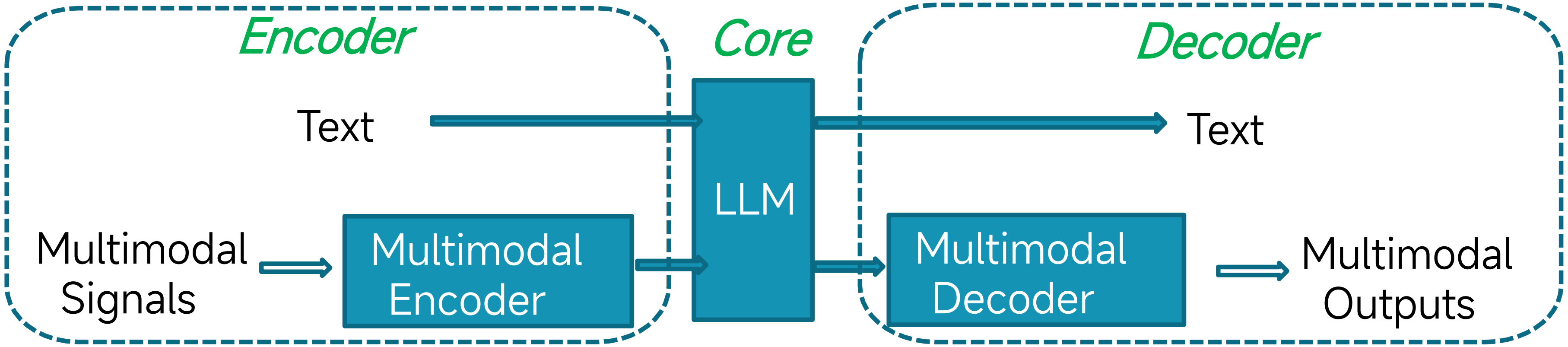
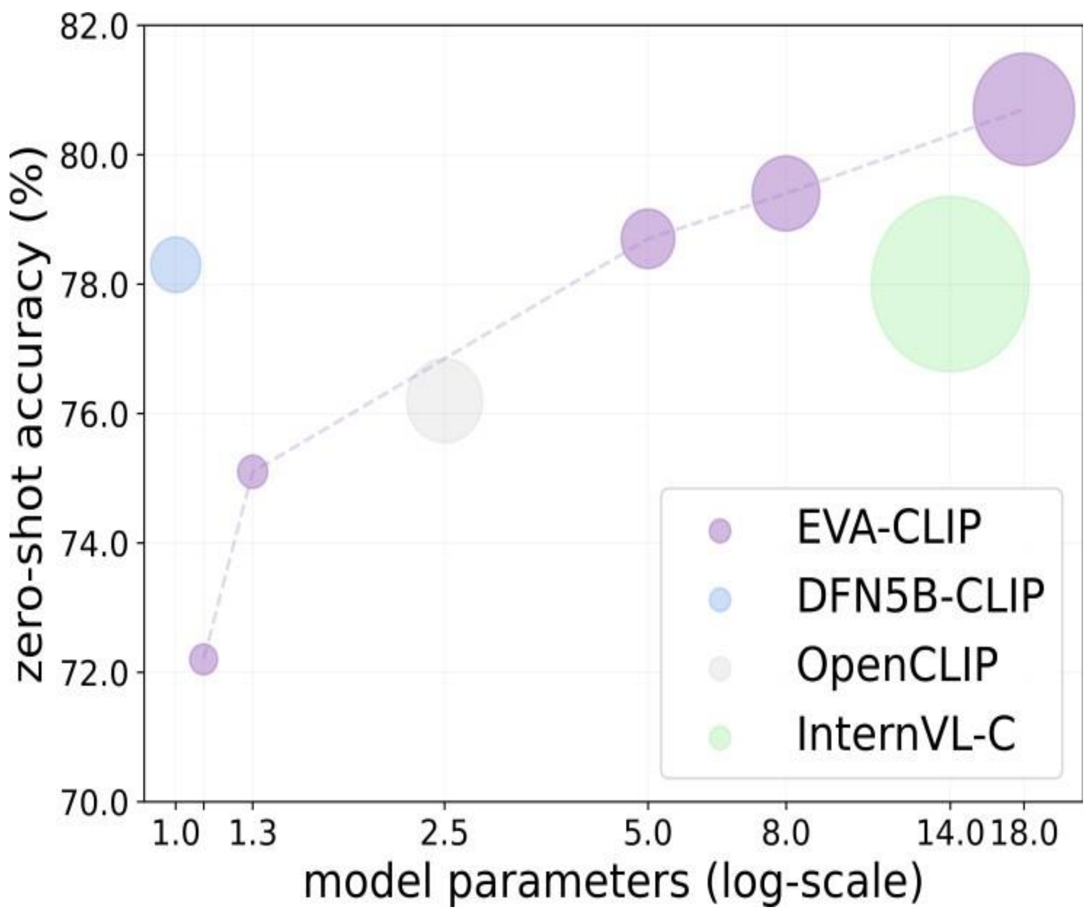
- Architecture-II: LLM as Joint Part of System
  - The role of the LLM is to perceive multimodal information, and react by itself, in a structure of Encoder-LLM-Decoder.
- Key feature:
  - LLM is the key joint part of the system, receiving multimodal information directly from outside, and delegating instruction to decoders/generators in a smoother manner.



Multimodal Encoding

Visual Encoder

- CLIP-ViT is the most popular choice for vision-language models.
    - Providing image representations well aligned with text space.
    - Scale well with respect to parameters and data.
- SigLIP is gaining increasing popularity (smaller and stronger)



[2303.15343] Sigmoid Loss for Language Image Pre-Training

Multimodal Encoding

Visual Encoder

- High-resolution Multimodal LLMs
  - Image slice-based: Split high-resolution images into slices
- Representatives:
  - GPT-4V, LLaVA-NeXT, MiniCPM-V 2.0/2.5, LLaVA-UHD, mPLUG-DocOwl 1.5, SPHINX, InternLM-XComposer2-4KHD, Monkey

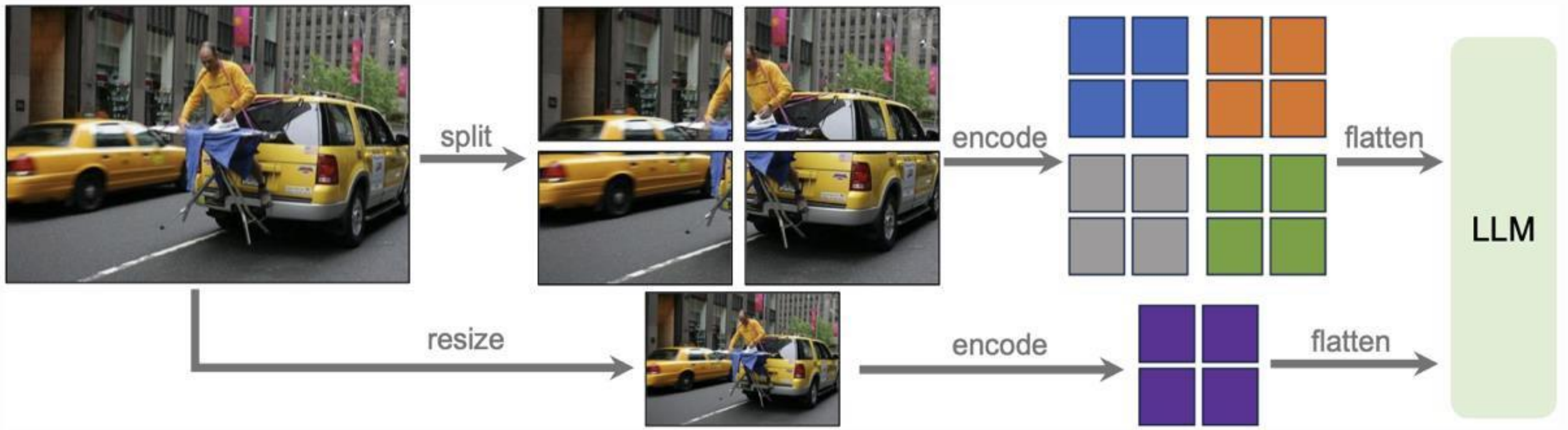
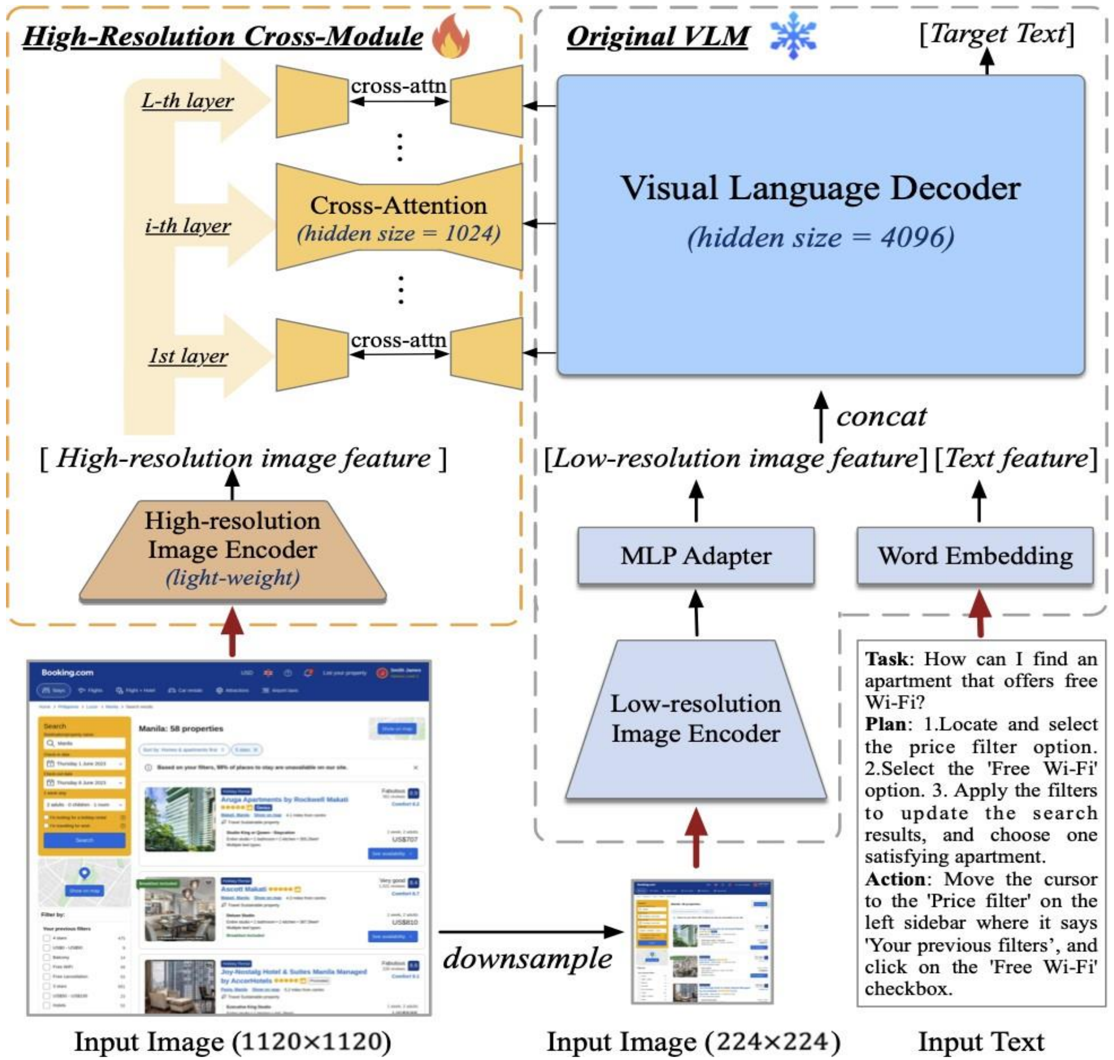


Illustration of dynamic high resolution scheme: a grid configuration of 2×2

Multimodal Encoding

Visual Encoder

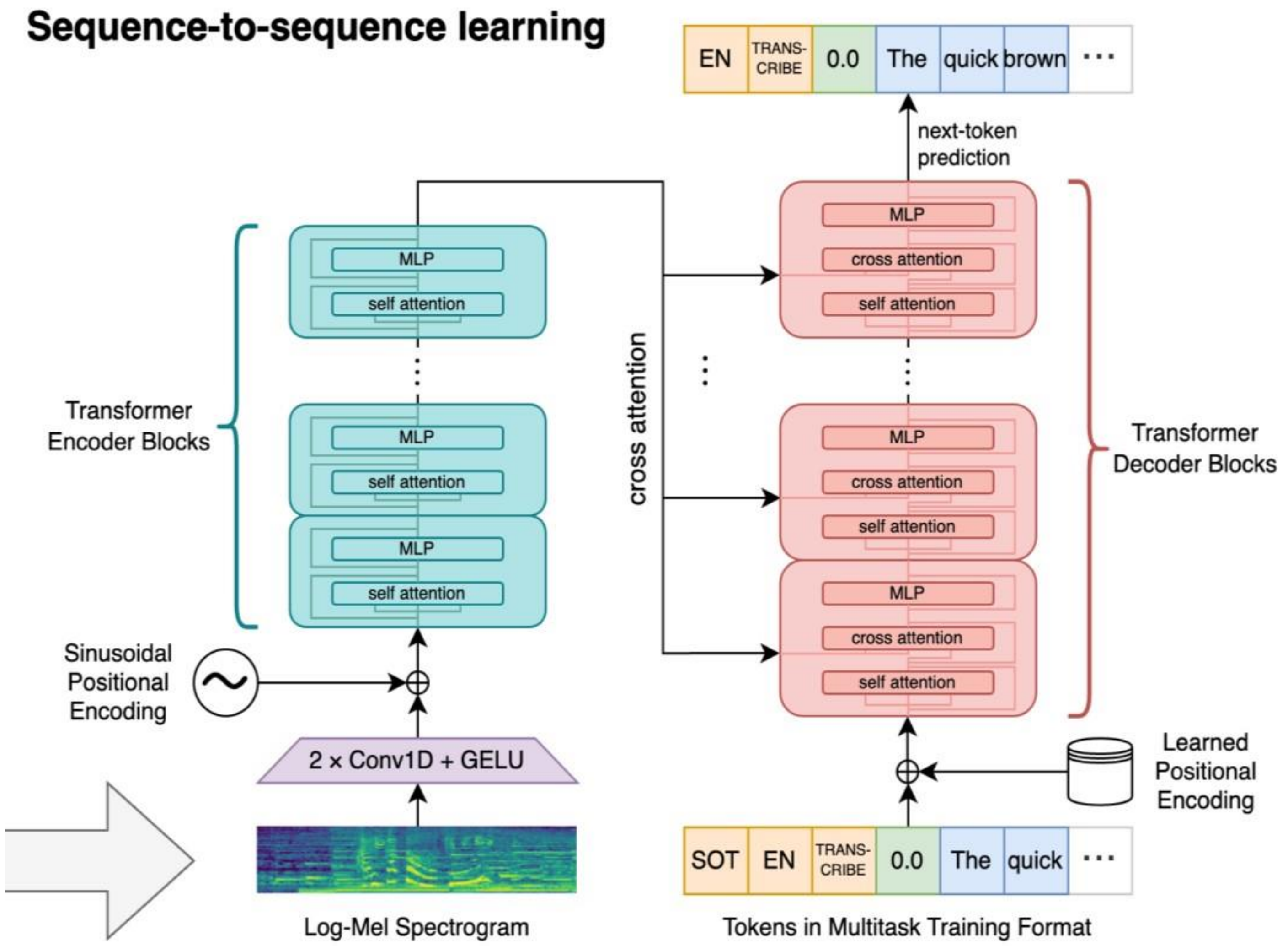
- High-resolution Multimodal LLMs
    - Dual branch encoders
- Representatives
    - CogAgent
    - Mini-Gemini
    - DeepSeek-VL
    - LLaVA-HR



Multimodal Encoding

Non-Visual Encoder

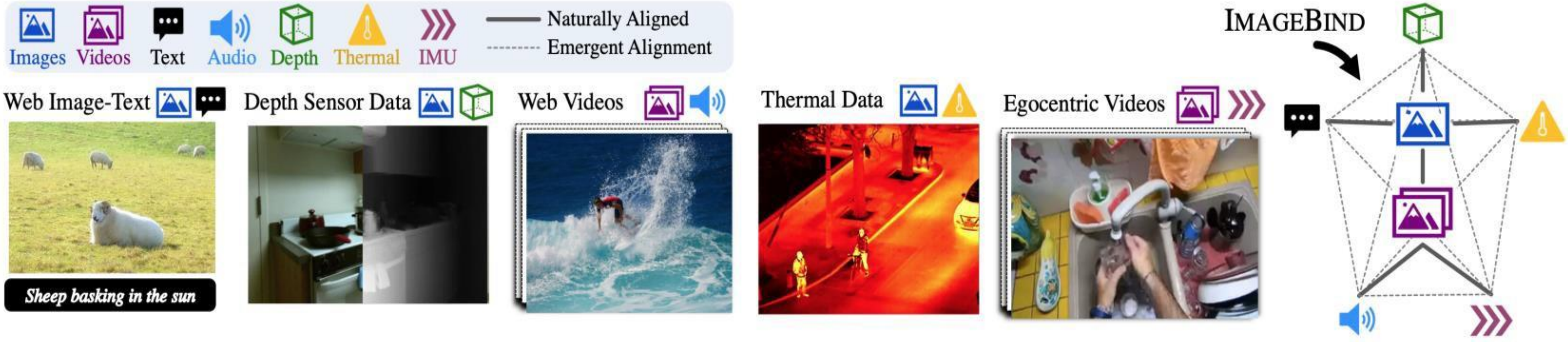
- Audio:
  - Whisper
  - AudioCLIP
  - HuBERT
  - BEATs
- 3D Point:
  - Point-BERT



Multimodal Encoding

Unified Multimodal Encoder

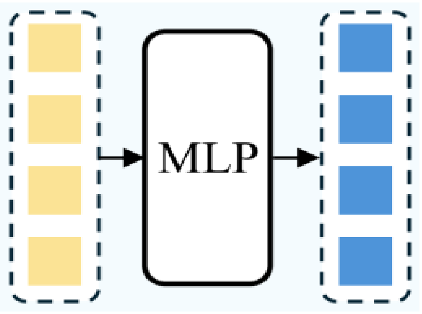
- ImageBind:
  - Embedding all modalities into a joint representation space of Image.
  - Well aligned modality representations can benefit LLM understanding

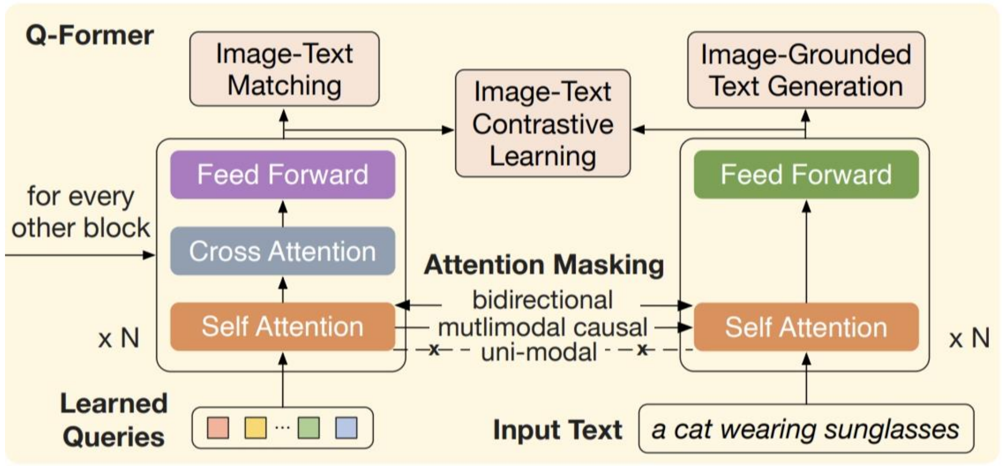


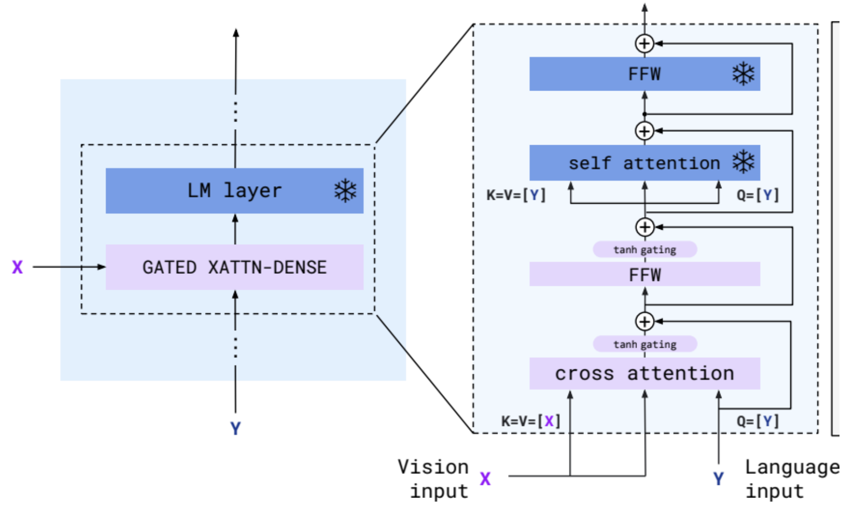
Input-side Projection

Methods to Connect Multimodal Representation with LLM

- Projecting multimodal (e.g., image) representations into LLM semantic space
  - Q-Former: BLIP-2, InstructBLIP, VisCPM, VisualGLM
  - Linear projection: LLaVA, MiniGPT-4, NExT-GPT
  - Two-layer MLP: LLaVA-1.5/NeXT, CogVLM, DeepSeek-VL, Yi-VL
  - Perceiver Resampler: Flamingo, Qwen-VL, MiniCPM-V, LLaVA-UHD



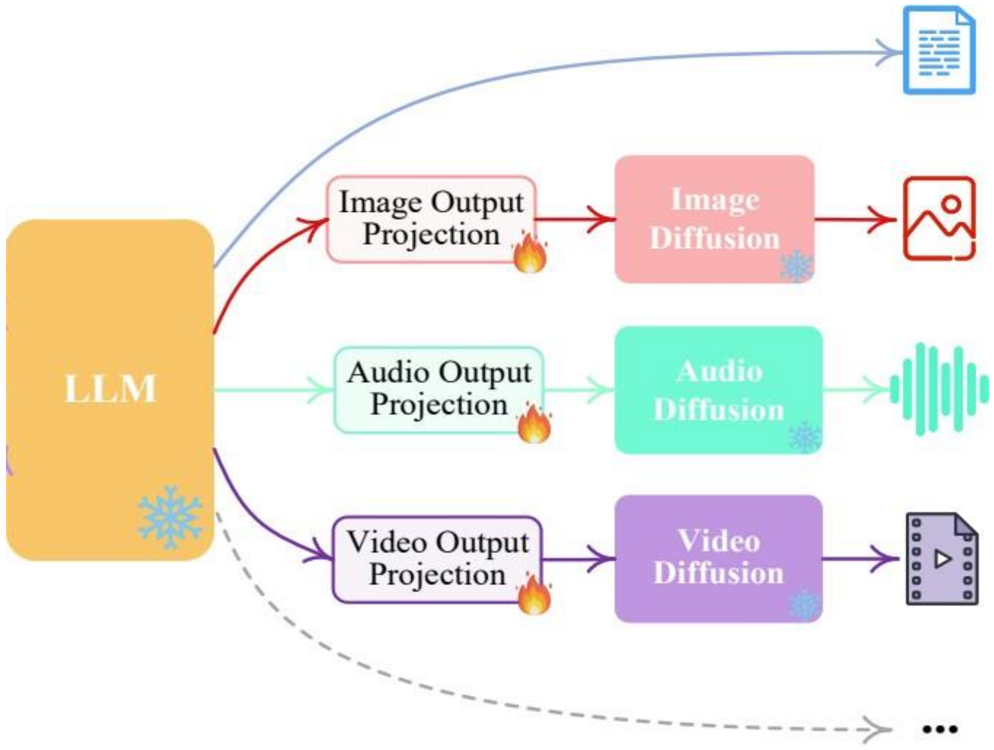




Multimodal Generation

Generation via Diffusion Models

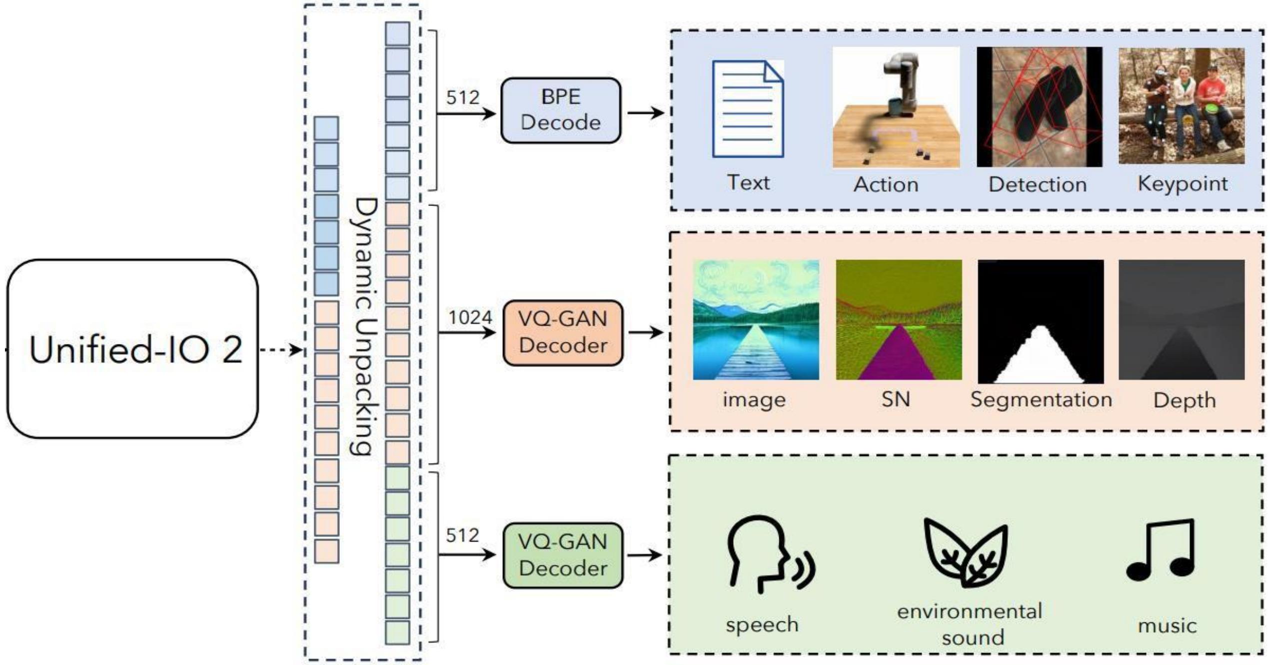
- Visual (Image/Video) Generator
  - Image Diffusion
  - Video Diffusion
- Audio Generator
  - Speech Diffusion
  - Audio Diffusion



Multimodal Generation

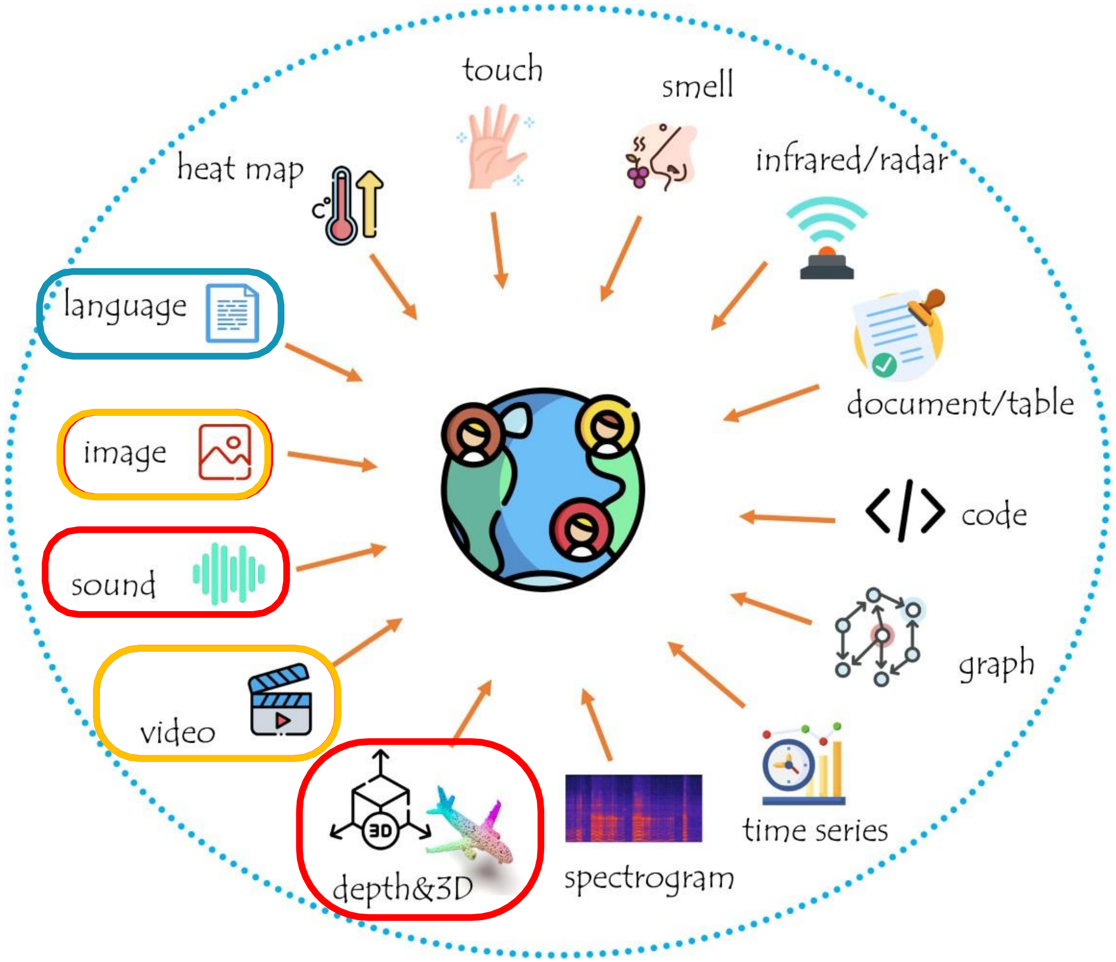
Generation via Codebooks

- Visual (Image/Video) Generator
  - VQ-VAE + Codebooks
  - VQ-GAN + Codebooks
- Audio Generator
  - SpeechTokenizer + Residual Vector Quantizer
  - SoundStream + Residual Vector Quantizer



Overview of Modality and Functionality

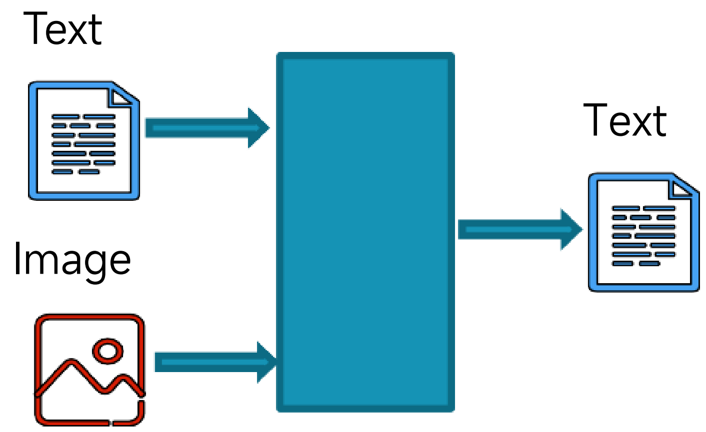
- Modalities



Language + Vision

Multimodal Perceiving

- Image-perceiving MLLM
  - Flamingo,
  - Kosmos-1,
  - Blip2, mPLUG-Owl,
  - Mini-GPT4, LLaVA,
  - InstructBLIP, Otter,
  - Chameleon,
  - Qwen-VL, GPT-4v,
  - …

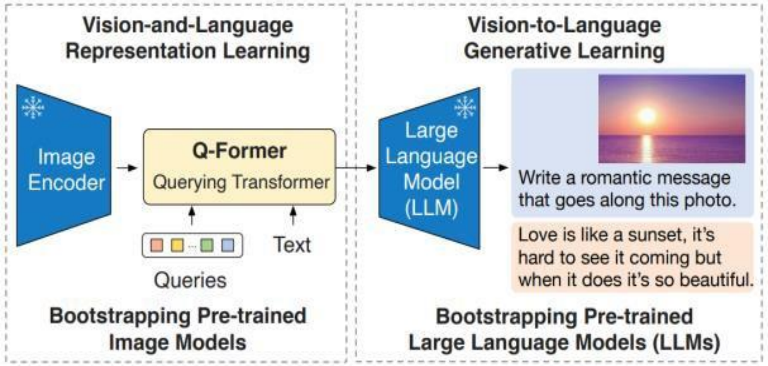


☞ Encode input images with external image encoders, generating LLM-understandable visual feature, which is then fed into the LLM. LLM then interprets the input images based on the input text instructions and produces a textual response.

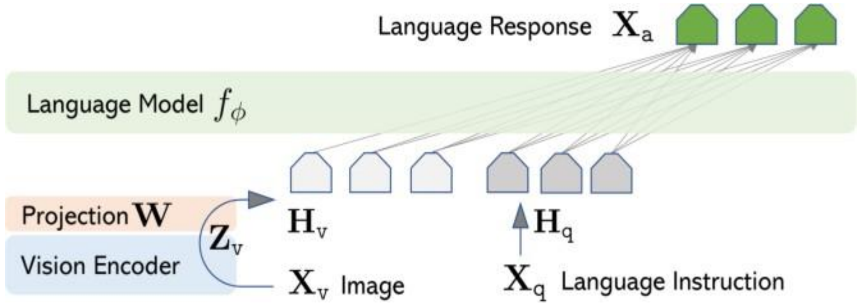
Multimodal Perceiving

- Image-perceiving MLLM

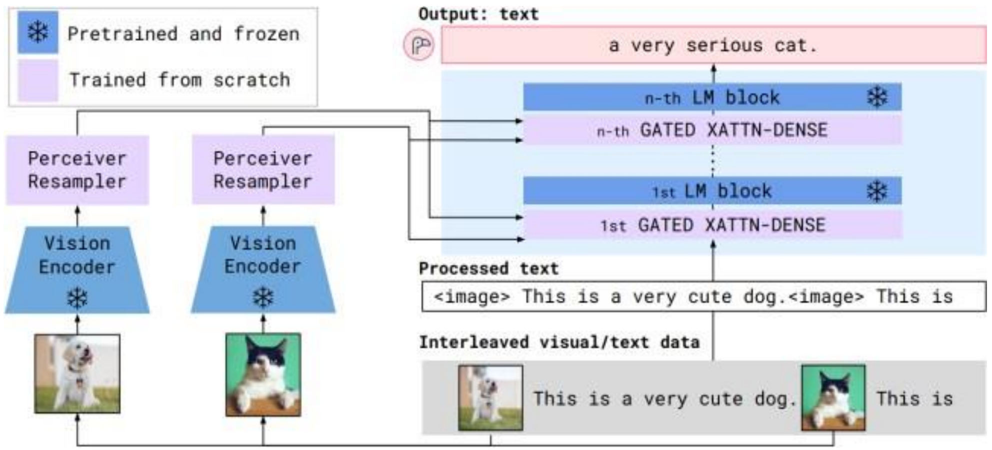
❖ Blip2



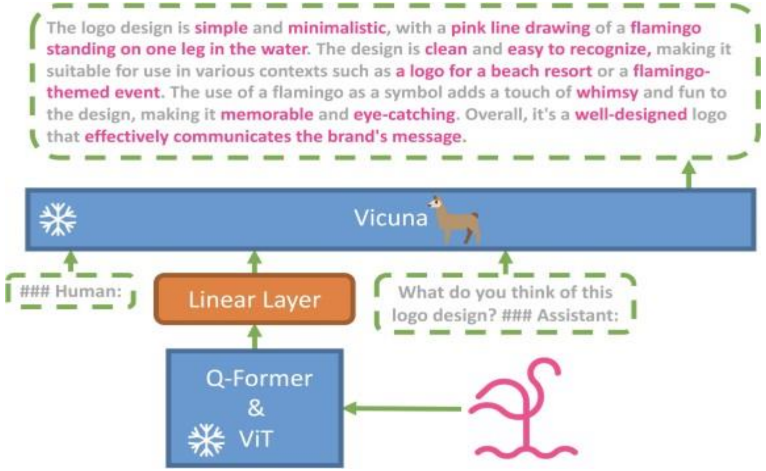
❖ LLaVA



❖ Flamingo

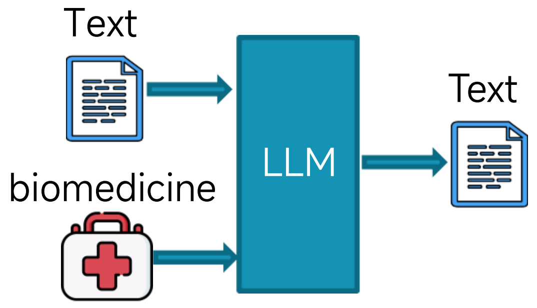


❖ Mini-GPT4



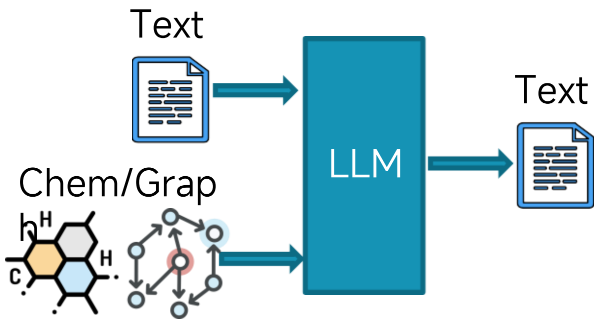
Multimodal Perceiving

- X-perceiving MLLM
  - Bio-/Medical & Healthcare
    - BioGPT
    - DrugGPT
    - BioMedLM
    - OphGLM
    - GatorTron
    - GatorTronGPT
    - MEDITRON
    - DoctorGLM
    - BianQue
    - ClinicalGPT
    - Qilin-Med
    - ChatDoctor
    - BenTsao
    - HuatuoGPT
    - MedAlpaca
    - AlpaCare
    - Zhongjing
    - PMC-LLaMA
    - CPLLM
    - MedPaLM 2
    - BioMedGPT



Multimodal Perceiving

- X-perceiving MLLM
  - Molecule & Chemistry
    - ChemGPT
    - SPT
    - T5 Chem
    - ChemLLM
    - MolCA
    - MolXPT
    - MolSTM
    - GIMLET
    - …
  - Graph
    - StructGPT
    - GPT4Graph
    - GraphGPT
    - LLaGA
    - HiGPT
    - …
  - Geographical Information System (GIS)
    - GeoGPT



Unified MLLM: Perceiving + Generation

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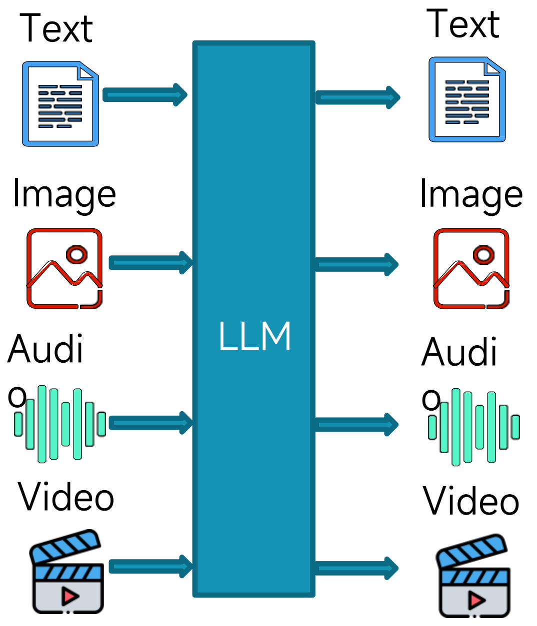
- Scenarios
  - Often, MLLMs need to not only understand the input multimodal information, but also to generate information in that modality.
  - Image Captioning
  - Visual Question Answering
  - Text-to-Vision Synthesis
  - Vision-to-Vision Translation
  - Scene Text Recognition
  - Scene Text Inpainting
  - …

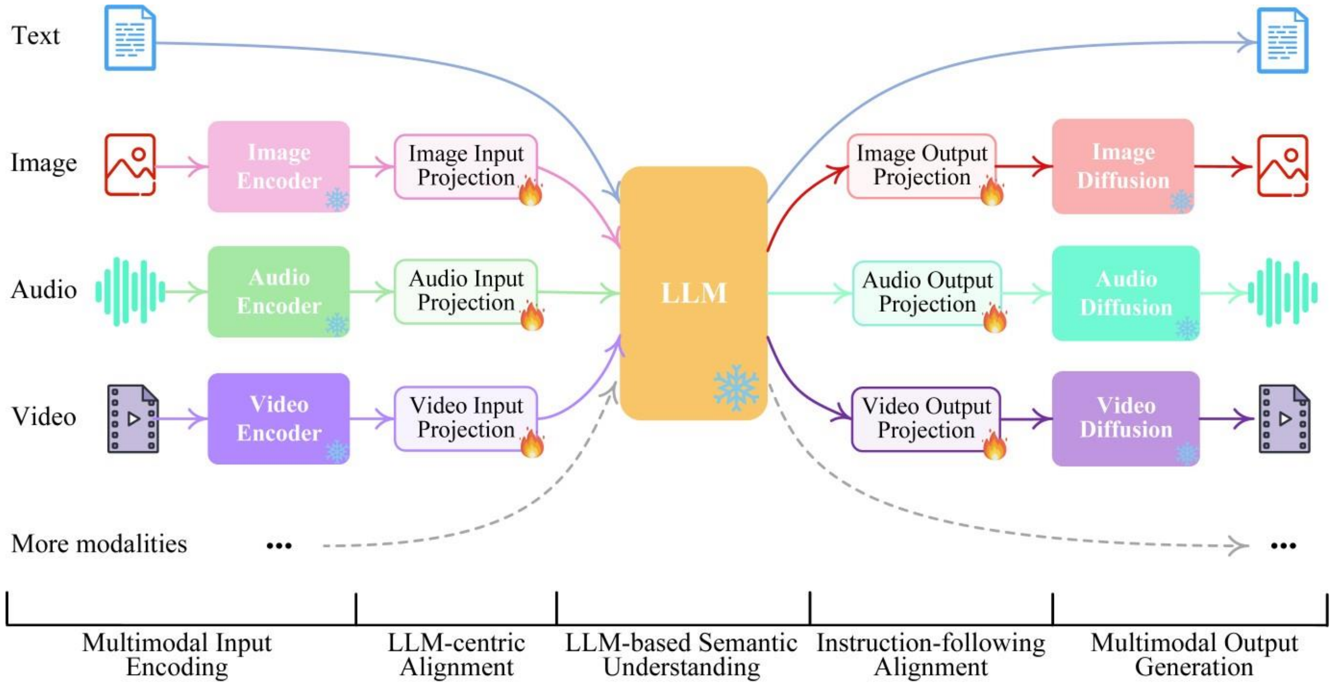
Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT
  - Unified-IO 2 (w/o video)
  - AnyGPT (w/o video)
  - CoDi-2
  - Modaverse
  - …



☞ Central LLMs take as input texts, audio, image and video, and freely generate texts, audio, image and video, or combination.

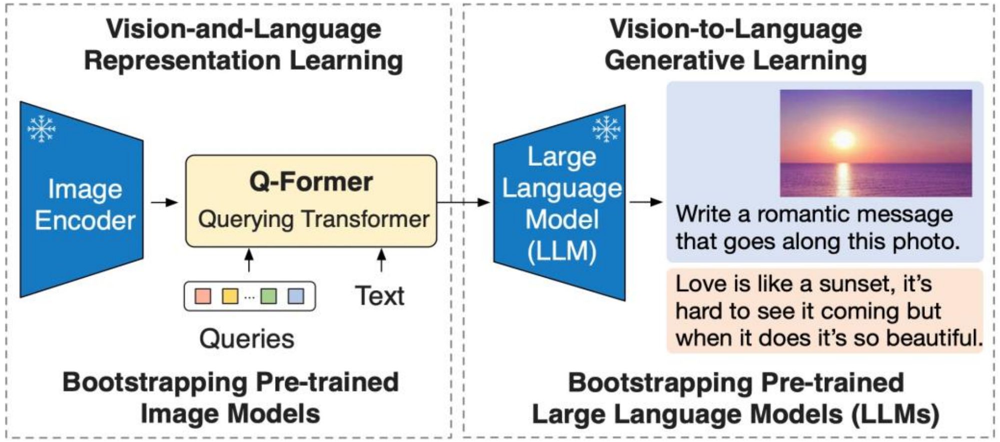
- Any-to-Any MLLM
  - NExT-GPT

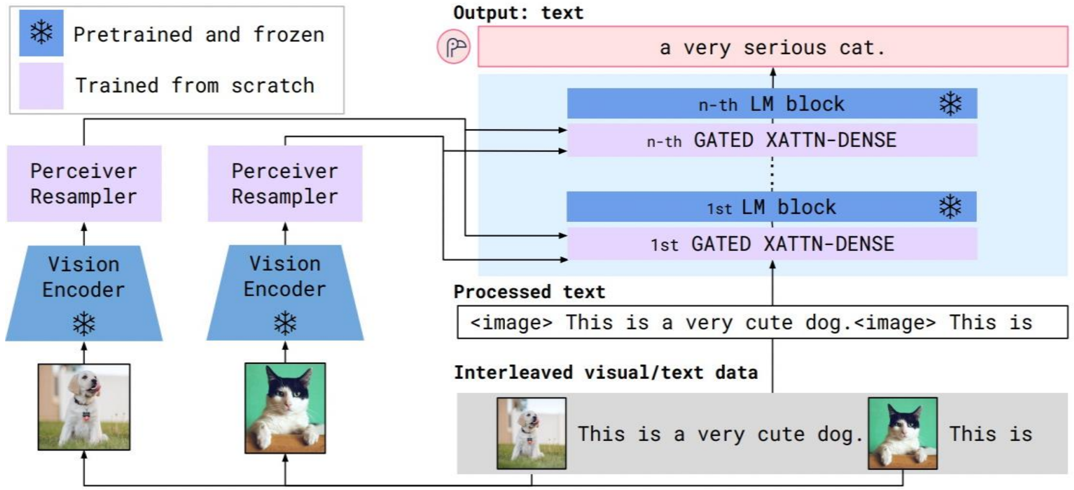


[2309.05519] NExT-GPT: Any-to-Any Multimodal LLM

Why Multimodal Instruction Tuning?

- Pretrained models aligns multiple modalities, can understand basic information from different modalities, and sometimes perform simple question-answering.
- Cannot follow complex instructions and often require task-specific fine-tuning for it to perform well on downstream tasks.





Why Multimodal Instruction Tuning?

- Multitask learning (with task tokens)

Training

☞ INPUT: <image><tok_task_1=short_cap>
OUTPUT: <generated short descriptions>

INPUT: <image><tok_task_2=yes_no>
OUTPUT: yes/no

Testing

Only with <tok_task_1>, <tok_task_2>…

Does not work with **<new_task=long_cap>**

- Instruction tuning (with natural language task instructions)

Training

☞ INPUT: <image>Describe this image briefly.
OUTPUT: <generated short descriptions>

INPUT: <image>Is this xxx?
OUTPUT: yes/no

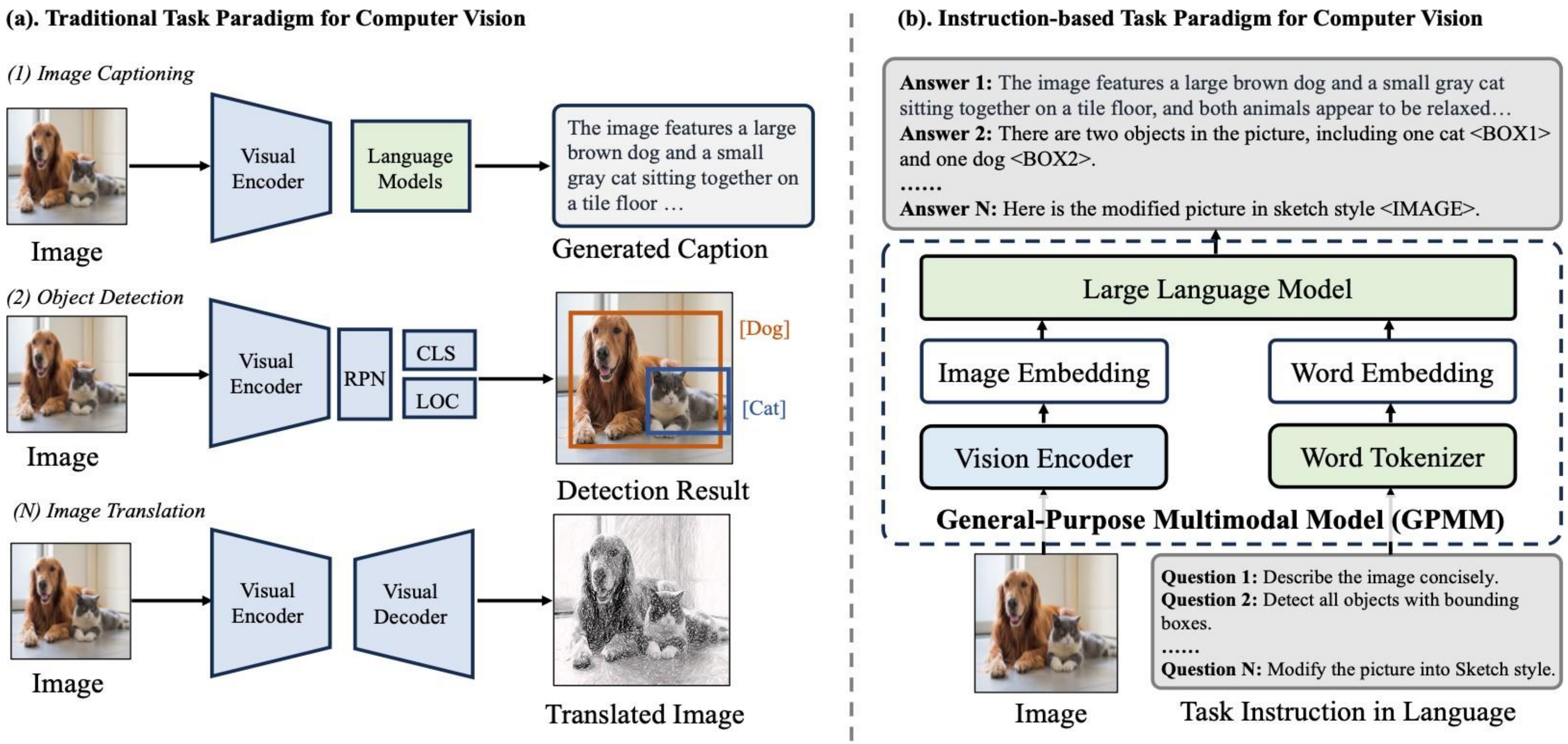
Testing

INPUT: <image>Describe this image in detail. OUTPUT: <long descriptions>

Generalizes to new instructions zero-shot.

Why Multimodal Instruction Tuning?

- From Single-Purpose to General-Purpose

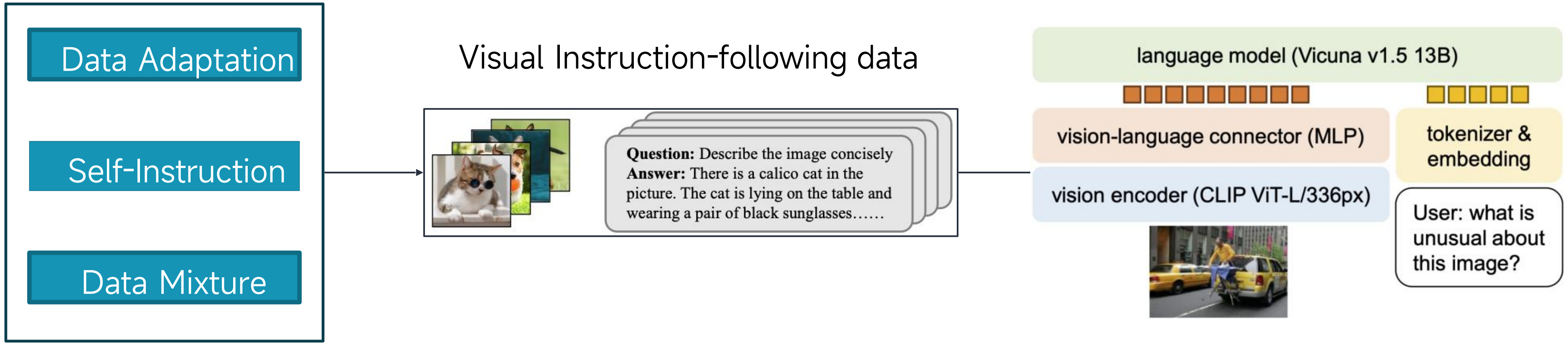


MLLM Instruction Tuning Framework

- Popular MLLMs:
  - LLaVA, MiniGPT4, LLaVA-NeXT, ViP-LLaVA, LLaVA-UHD, MiniCPM, Qwen-VL, CogAgent, InternVL, mPLUG-OWL, Monkey, MiniGemini, LLaVA-HR, SPHINX, DeepSeek-VL, MoAI

Data Construction

Visual Instruction Tuning Framework Example: LLaVA-1.5



Training Paradigms

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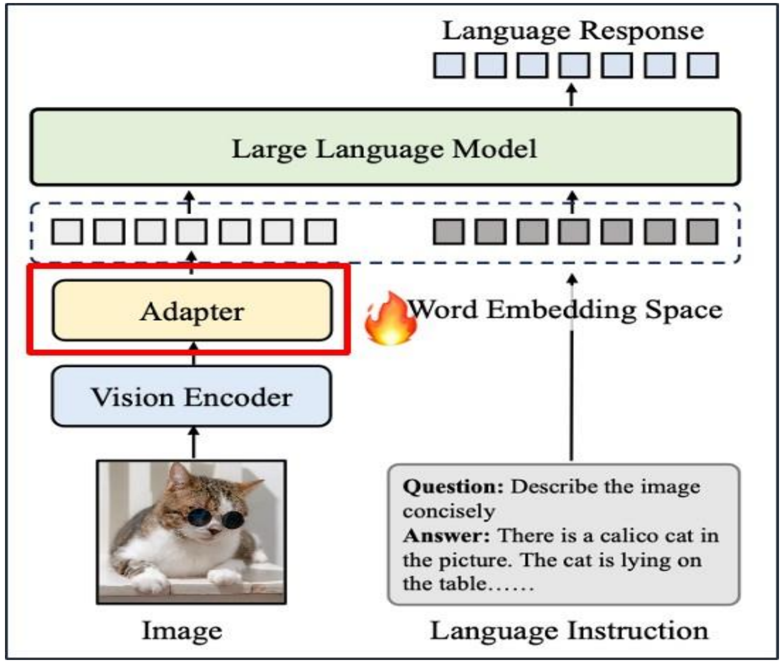
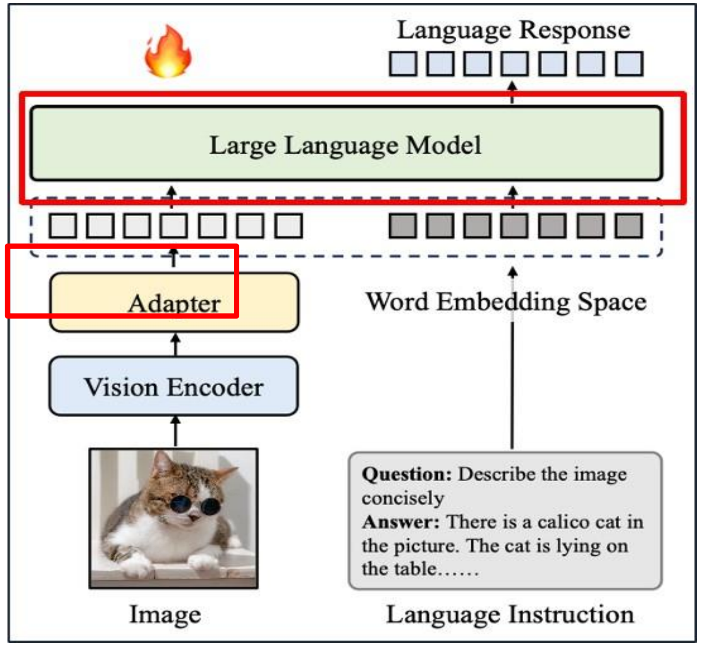
Stage1: Pretraining Stage

- Align different modalities, provide world knowledge



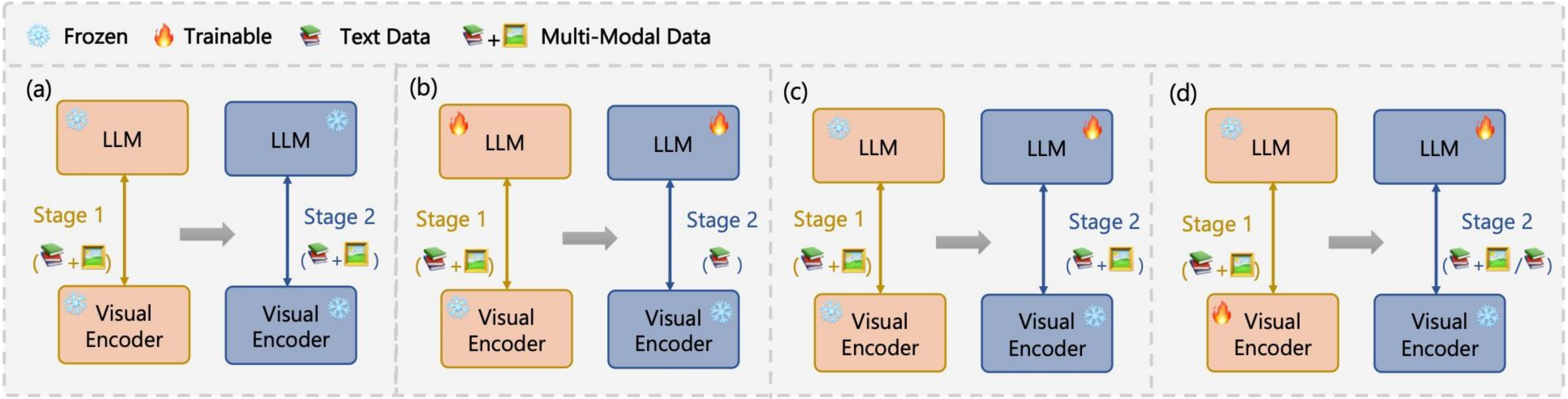
Stage2: Instruction Tuning Stage

- Teach models to better understand the instructions from users and fulfill the demanded tasks.



Training Paradigms

Training paradigms of popular multimodal large language models.



MiniGPT4 Kosmos-1 LLaVA mPLUG-Owl

Visual Instruction Tuning

- It introduces LLaVA: Large Language and Vision Assistant
  - LLaVA is an end-to-end trained LMM that connects a vision encoder and an LLM
  - It is instruction tuned on multimodal language-image instruction-following data.
- Text-only GPT-4 is used

Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg

User: What is unusual about this image?

LLaVA: The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

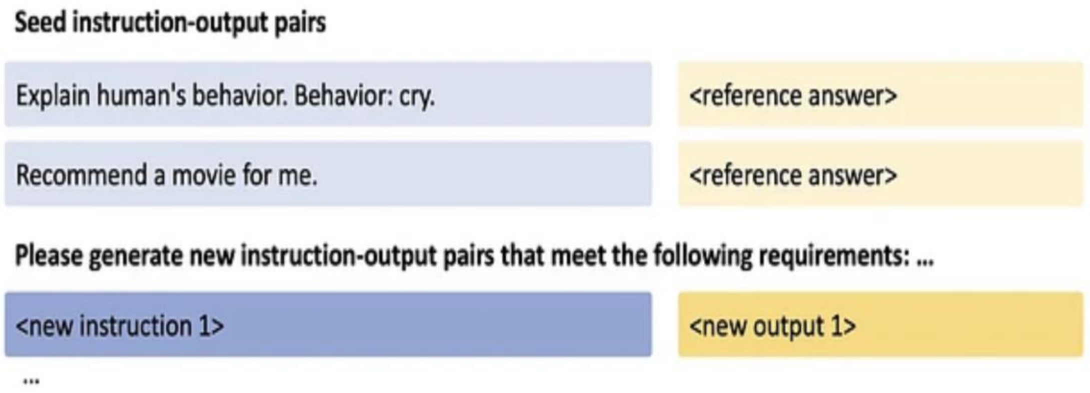
[2304.08485] Visual Instruction Tuning

Instruction Tuning in LLMs

➤ How to collect the instruction tuning data?

- Human-Human: high-quality, hand-written by humans – high cost
- Human-Machine: strong LLM-based teacher like ChatGPT – affordable cost

Using ChatGPT



What do we need?

- Strong pretrained vision and language models
- Cross-modal Connector
- Tuning the model for following multimodal instructions

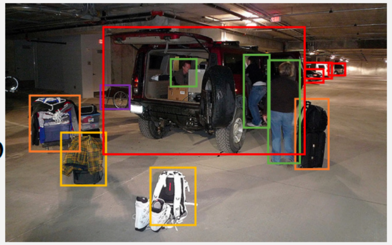
Creating the Dataset

LLaVA-Instruct-158K

- Conversation: 58K
- Detailed description: 23K
- Complex reasoning: 77K

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

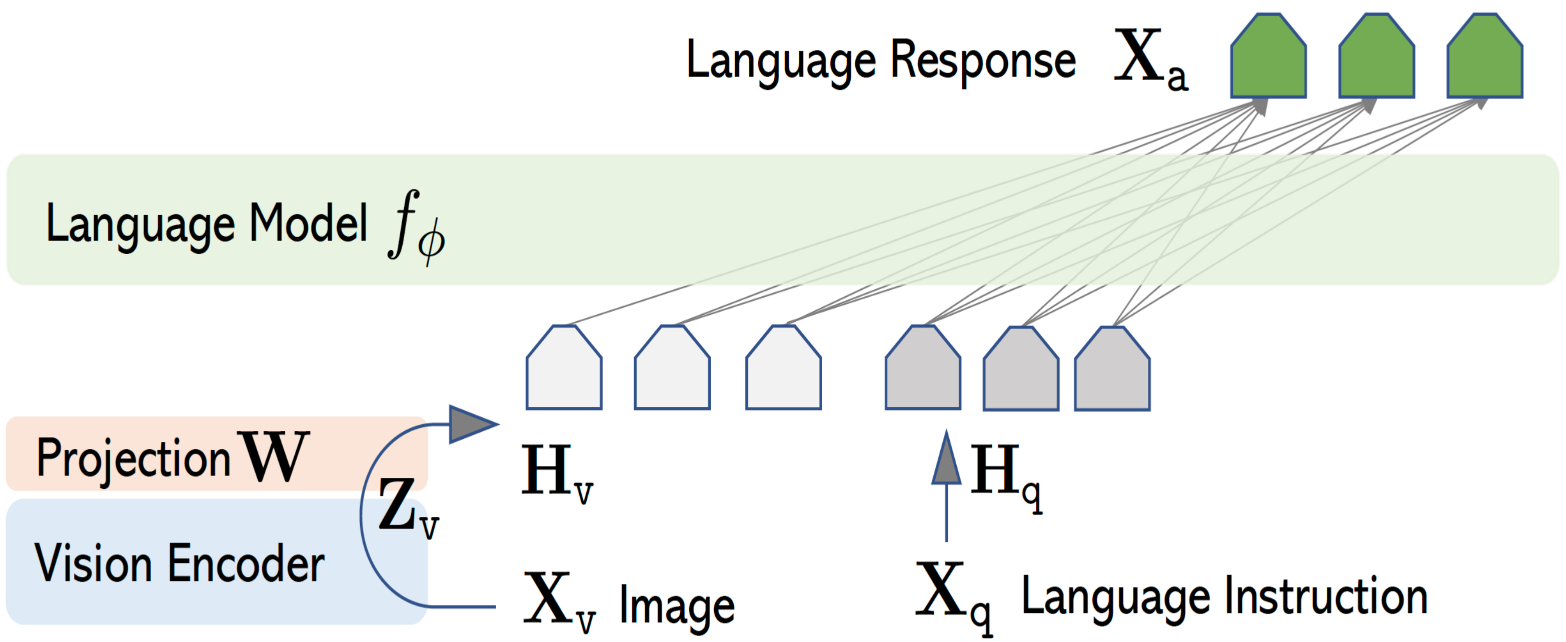
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

LLaVA model

Components

- **Vision Encoder:** CLIP-ViT-L/14
- **Projection:** Linear layer
- **Language Model:** Vicuna, LLaMA-2-Chat, MPT-Chat, etc.

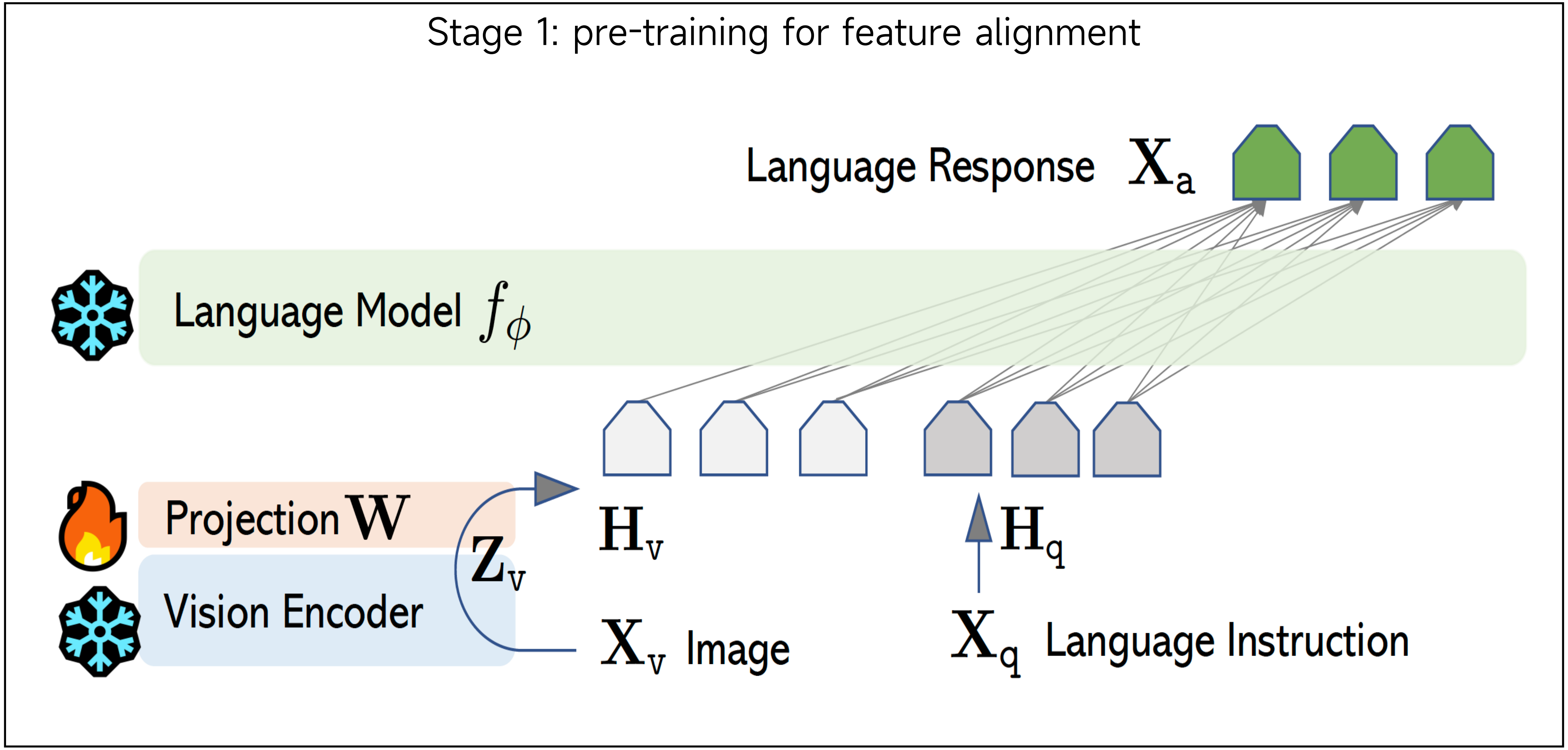
The architecture



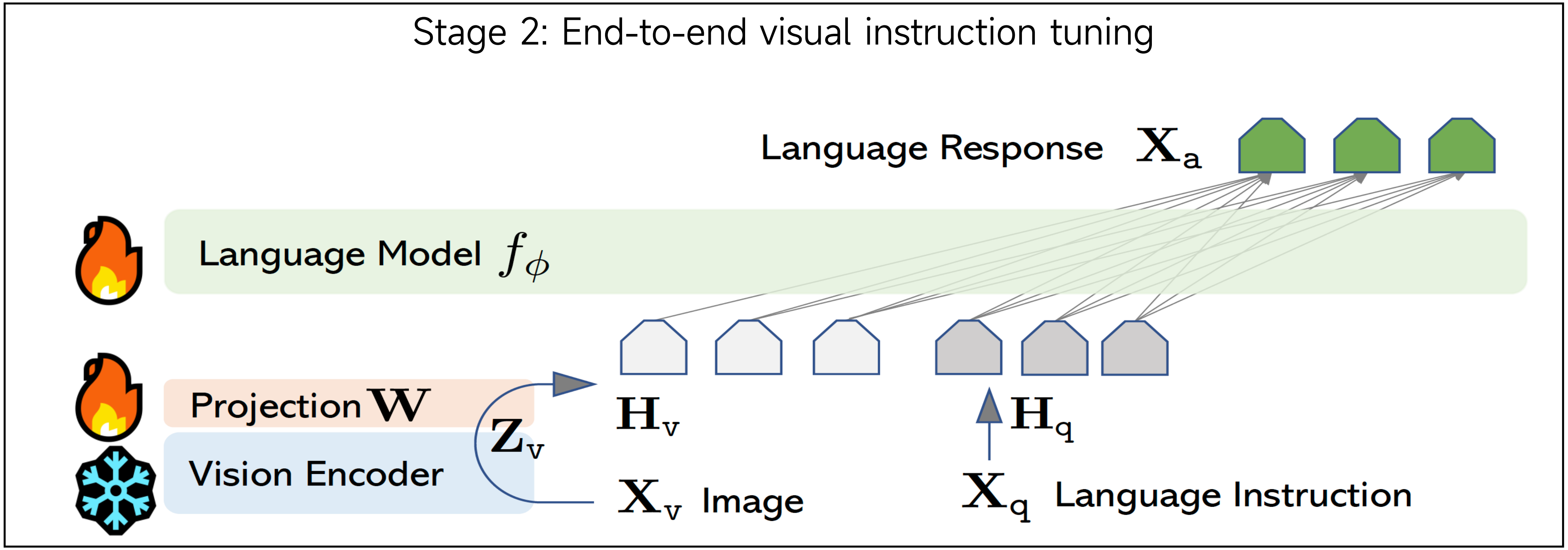
Training

Data

A filtered CC3M subset (595K)



Training



Two tasks

- Visual Chat: LLaVA-Instruct-158K for open-ended user-oriented visual task.
- Science QA: Multimodal reasoning dataset for the science domain

LLaVA-Bench (In-the-Wild)

南方科技大学 SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Goals of building this benchmark

- Challenging
  - knowledge beyond training data
  - Multilingual understanding
  - Perception of subtle details
- Evaluation
  - Consistency
  - Prompt robustness

Challenging examples from LLaVA-Bench (In-the-Wild):

		
	ICHIRAN Ramen [source]	Filled fridge [source]
Annotation	A close-up photo of a meal at ICHIRAN. The chashu ramen bowl with a spoon is placed in the center. The ramen is seasoned with chili sauce, chopped scallions, and served with two pieces of chashu. Chopsticks are placed to the right of the bowl, still in their paper wrap, not yet opened. The ramen is also served with nori on the left. On top, from left to right, the following sides are served: a bowl of orange spice (possibly garlic sauce), a plate of smoke-flavored stewed pork with chopped scallions, and a cup of matcha green tea.	An open refrigerator filled with a variety of food items. In the left part of the compartment, towards the front, there is a plastic box of strawberries with a small bag of baby carrots on top. Towards the back, there is a stack of sauce containers. In the middle part of the compartment, towards the front, there is a green plastic box, and there is an unidentified plastic bag placed on it. Towards the back, there is a carton of milk. In the right part of the compartment, towards the front, there is a box of blueberries with three yogurts stacked on top. The large bottle of yogurt is Fage non-fat yogurt, and one of the smaller cups is Fage blueberry yogurt. The brand and flavor of the other smaller cup are unknown. Towards the back, there is a container with an unknown content.
Question 1	What's the name of the restaurant?	What is the brand of the blueberry-flavored yogurt?
Question 2	Describe this photo in detail.	Is there strawberry-flavored yogurt in the fridge?

Thank You